C-C-91d
July 1, 1968
SUPERSEDING
Fed. Spec C-C-91c
September 18, 1962

FEDERAL SPECIFICATION

CANDLE, ILLUMINATING

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers two types of candles suitable for illuminating purposes (see 6.1).

1.2 Classification.

1.2.1 Types. The candles furnished under this specification shall be of the following types, as specified (see 6.2):

Type I - 8 hour.
Type II - 6 hour.

2. APPLICABLE DOCUMENTS

2.1 Specification and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Box, Fiberboard.
PPP-B-676 - Boxes, Setup.
PPP-F-320 - Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes.
PPP-T-45 - Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks, as outlined under General Information in the Index of Federal Specifications and Standards, and at the prices

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indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Forth Worth, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks, and the Index of Federal Specifications and Standards, from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. The manufacturer is permitted wide latitude in the selection of raw materials provided the candles conform to the requirements of this specification.

3.1.1 Wick. The wick of the candles shall be satisfactory quality braided or plaited cotton yarn to assure meeting the requirements specified in 3.9.

3.2 Solubility. The candle material shall dissolve in carbon tetrachloride with no more than a trace of residue when tested as specified in 4.3.2.

3.3 Appearance and construction. The candles shall be cylindrical with a shoulder and shall have a smooth finish. The wick shall be properly centered and shall not be visible along the shaft of the candle. The wick shall be visible on the bottom of the candle and shall extend the entire length of the candle and beyond the tip.

3.4 Candle weight and dimensions. Six type I candles or eight type II candles shall weigh a total of not less than 14.5 ounces nor more than 16.5 ounces. The candles, as measured from the butt to the shoulder, shall not exceed 8-1/2 inches in length and the diameter of the candle shall be not less than 3/4 inch nor more than 15/16 inch.

3.5 Color. The candles shall be uniformly white in color.

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3.6 Odor. The finished product shall be free of unpleasant odor (pungent, unsavory) during burning and nonburning.

3.7 Deflection. The candles shall show a deflection of not more than 3 angular degrees, when tested as specified in 4.3.2.

3.8 Deformation. When tested as specified in 4.3.2 the candles shall be easily separated and shall not deform to the extent that the performance of the candles is affected as specified in 3.9.

3.9 Burning time and quality. Type I candles shall have a minimum burning time of 7-1/2 hours and type II candles shall have a minimum burning time of 5-1/2 hours. The candles shall burn uniformly with an even flame. The flame shall consume the candle and wick without dripping, excessive smoking objectionable odor, and without the necessity of trimming the wick when tested as specified in 4.3.2.

3.10 Workmanship. The finished product shall be clean, free from dirt and foreign matter, uniform, free of fractures (breaks) and distortions (warped, sags, runs, two or more fused together).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that suppliers and services conform to prescribed requirements.

4.2 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except when otherwise indicated hereinafter.

4.2.1 Inspection of materials and components. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended or qualified in this specification or applicable purchase documents.

4.3 Inspection of the end item.

4.3.1 Examination of the end item. The end item shall be examined for the defects in the applicable sub-paragraphs of the inspection level and acceptable quality levels (AQLs) set forth in 4.3.1.6. The lot size shall be expressed in filled unit containers of the same type and size, containing candles of the same type for the examination in 4.3.1.1, 4.3.1.2, 4.3.1.3, and 4.3.1.4, and in units of shipping containers for the examination in 4.3.1.5.

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4.3.1.1 Examination of the candle for defects in appearance, construction, workmanship, odor and color. The sample unit for this examination shall be one candle. No more than two candles shall be drawn from any one unit container.

Examine	Defect
Appearance and construction	Not cylindrical with a shoulder. Not smoothly finished. Wick not braided, or plaited, missing, or not as specified; not centered in candle; visible along the shaft of the candle; not visible at bottom of candle; does not extend entire length of candle and beyond the tip.
Workmanship	Not clean; contains dirt, dust, or foreign matter Not uniform. Fractures (breaks). Distortions (warped, sags, runs or two or more fused together).
Odor	Unpleasant odor (nonburning).
Color	Not uniformly white.

4.3.1.2 Examination of the end item for defects in weight. The sample unit shall be six candles for type I and eight candles for type II, taken from one package.

Examine	Defect
Weight of 6 candles (type I)	More than 16.5 ounces or less than 14.5 ounces.
Weight of 8 candles (type II)	More than 16.5 ounces or less than 14.5 ounces.

4.3.1.3 Examination of the end item for defects in dimensions. The sample unit for this examination shall be one candle.

Examine	Defect
Types I and II	
Length (butt to shoulder)	More than 8-1/2 inches.
Diameter	More than 15/16 inch or less than 3/4 inches.

4.3.1.4 Examination for average net content. The sample unit for this examination shall be one filled unit container. The lot shall be unacceptable if the average net content per unit container is less than specified.

4.3.1.5 Examination of preparation for delivery. An examination shall be made to determine that the packaging, packing and markings comply with

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section 5 of this specification. The sample unit shall be one shipping container fully packed, selected just prior to the closing operation. Closed shipping container fully prepared for delivery shall be examined for closure defects.

Examine	Defect
Packaging (as applicable)	Packaging material not as specified. Closure not completed or as specified. Box not secured with tape, as specified. Separator paper missing, not interleaved between the candles as specified.
Packing (as applicable)	Number of interior packages per shipping container not in accordance with contract requirements. Bulged or distorted container. Container not as specified.
Markings (exterior and interior)	Omitted, illegible, incorrect, incomplete.

4.3.1.6 Inspection levels and acceptable quality levels (AQLs) for examination. The inspection levels and AQLs, expressed in defects per 100 units, shall be as follows:

Examination paragraph	Inspection level	AQLs
4.3.1.1	I	4.0
4.3.1.2	S-2	4.0
4.3.1.3	S-2	4.0
4.3.1.4	S-2	N.A.
4.3.1.5	S-2	4.0

4.3.2 Testing of the end item. The methods of testing listed in table II shall be followed for each lot. For purposes of sampling, the lot shall be expressed in the number of candles of the same type manufactured as a single batch. The sample unit for testing shall be six candles randomly selected throughout the lot. The number of sample units shall be as indicated in table I. All test reports shall contain the individual values utilized in expressing the final result. The lot shall be unacceptable if one or more sample units or the lot average fail to meet any test requirement specified.

TABLE I. Sampling for end item testing

Lot size (candles)	Number of sample units
800 or less.	2
801 up to and including 22,000.	3
22,001 or more.	5

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TABLE II. Instructions for testing the end item.

CHARACTERISTIC	Specification Reference		Requirements Applicable To		Number Determinations per sample unit	Results Reported As		Inspect Level	AQL
	Requirement	Test Method	Individ Unit	Lot Aver		Pass or Fail	Numerically to Nearest		
Temperature deflection	3.7	4.4.1		X	1		1°angular		
Deformation	3.8	4.4.2	X		1	X			
Solubility	3.2	4.4.3	X		1	X			
Burning 1/									
Quality	3.9	4.4.4	X		1	X			
Time (minimum)	3.9	4.4.4	X		1	X			

1/ All burning characteristics shall be observed on the same candle.

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4.4 Test procedures.

4.4.1 Deflection.

4.4.1.1 Apparatus.

4.4.1.1.1 Deflection tester. The deflection tester shall consist of an accurate protractor chart marked off at each 10 degrees, with the first 10 degrees subdivided into 10 equal parts. A suitable clamp arrangement shall be used to hold candle and protractor chart (see figure 1). The protractor chart shall be in a vertical position.

4.4.1.1.2 Constant temperature cabinet. The constant temperature cabinet shall be large enough to hold the deflection tester assembly and shall be capable of maintaing a temperature of 38° ± 1.0°C. (100.4° ± 1.8°F) throughout the cabinet continously for 24 hours.

4.4.1.2 Procedure. The candle shall be placed in the deflection tester with the length of the candle (excluding tip) 5.5 inches above the top of the clamp. The candle shall be at an angle of 5 degrees from the vertical and so placed than an angular deflection of the specimen can be accurately noted on the protractory chart. The assembly shall be placed in the constant temperature cabinet maintained at 38° ± 1.0°C. (100.4° ± 1.8°F.). Allow to remain at this temperature for 24 hours with periodic observations for possible failure. Record angular deflection. The angular deflection shall be considered the increase from the initial 5-degree position.

4.4.2 Deformation. Four candles (two on two) shall be placed in a rigid box with the wicks in a horizontal position. The box shall be just wide enough for two candles (see figure 2). A weighed flat board shall be placed on the top layer of candles so that 4 ± 0.2 pounds are evenly distributed over the top two candles. The assembly with the candles shall be placed in a constant temperature cabinet for 24 hours at a temperature of 38° ± 1.0°C. (100.4° ± 1.8°F.). Record the deformation of the candles.

4.4.3 Solubility. Place a candle in a 250-milliliter beaker and melt carefully at a low temperature. Stir the molten material. Pour off a portion of about 5 grams into a 250-milliliter Erlenmeyer flask. Allow to cool. Add 100 milliliters of carbon tetrachloride and heat until the candle material is dissolved. No more than a trace of insoluble material shall remain.

4.4.4 Burning time and quality. The burning time of the candles shall be determined at an ambient temperature of 20° to 25°C. under conditions free from drafts but with access to air. Record the burning time and burning qualities of the candles, including odor.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B or C as specified (see 6.2).

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5.1.1 Level A. Twelve candles of one type only shall be packaged in a snug-fitting folding paperboard box conforming to variety 1, style VIII, type B, class a of PPP-B-566; or setup paperboard box conforming to type I, variety 1, class A, style 2 of PPP-B-676. The candles shall be interleaved or individually wrapped with a material that is compatible with the candles, in a manner to prevent candle-to-candle contact or candle contact with the inner faces of the box. Box closure shall be secured with 2-inch minimum width gummed paper tape conforming to type III, grade B of PPP-T-45, applied at the center of the length opening and extending along the bottom and up each side.

5.1.2 Level B (civil agencies). For civil agency procurements the requirements in 5.1.1 shall apply.

5.1.3 Level C. Candles shall be packaged to afford adequate protection against physical damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets this requirement.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Two hundred and eighty-eight candles of one type only, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, grade V2s of PPP-B-636. Each shipping container shall be closed, waterproofed and reinforced with flat strapping or tape banding in accordance with the appendix of the container specification.

5.2.2 Level B. Two hundred and eighty-eight candles of one type only, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. Each shipping container shall be closed in accordance with method II as specified in the appendix of the container specification.

5.2.2.1 When specified (see 6.2), the shipping container shall be a grade V3c or V3s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with the appendix of the box specification. The shipping container material may also be grade V4s of PPP-F-320.

5.2.3 Level C. Candles, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery to destination at the lowest transportation rate for such supplies. Containers shall be in accordance with rules or regulations of carriers applicable to the mode of transportation.

5.3 Marking.

5.3.1 Civil agencies. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123, and shall include the following legend:

"STORE IN A COOL PLACE -
DO NOT EXPOSE TO DIRECT SUNLIGHT".

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5.3.2 Military agencies. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129, and shall include the following legend:

"STORE IN A COOL PLACE -
DO NOT EXPOSE TO DIRECT SUNLIGHT".

6. NOTES

6.1 Intended use. The candles covered by this specification are intended for emergency illumination when other forms of lighting are unavailable.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type required (see 1.2.1).
(c) Selection of the applicable levels of packaging and packing (see 5.1 and 5.2).
(d) When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).

MILITARY CUSTODIANS:

Army - GL
Air Force - Coordination waived

Review activity:

Army - MD

User activities:

Army - EL, ME
Navy - MC

Preparing activity:

Army - GL

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HEW
JUS
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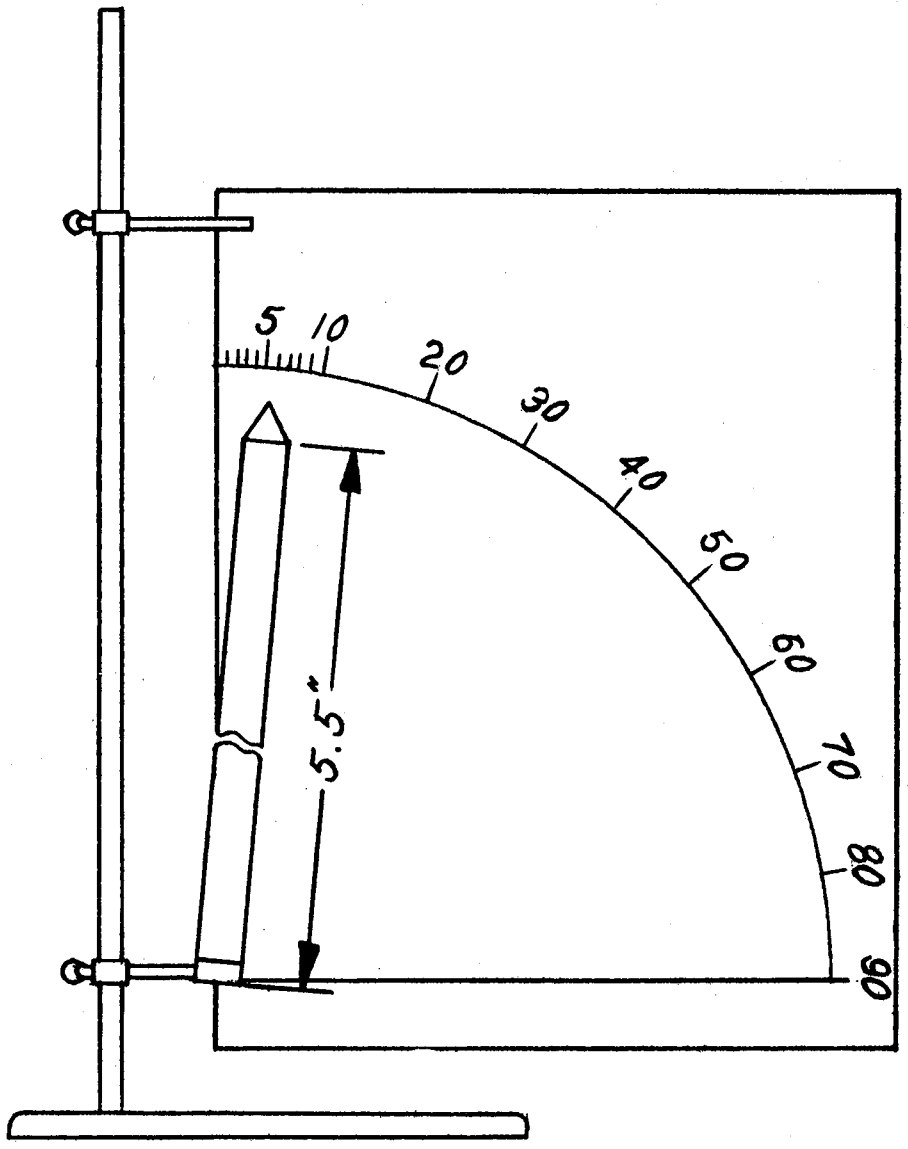


FIGURE 1. Deflection tester.

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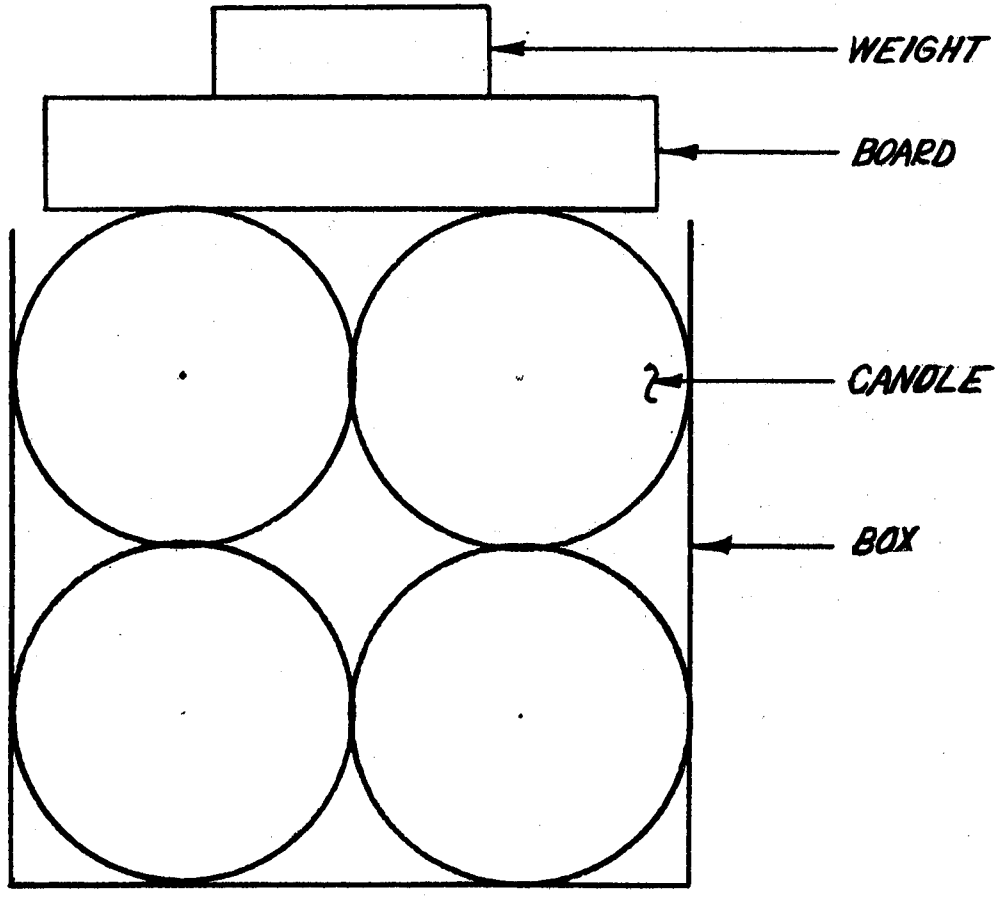


FIGURE 2. Deformation tester.

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 10 cents each.

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SPECIFICATION ANALYSIS SHEET

Form Approved
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INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

Candle, Illuminating - C-C-91d

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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